

Vidya Sagar Sharma

Institute Postdoctoral Fellow, IIT Madras

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Research Profile

Algorithms, graph theory, combinatorial optimization, computational complexity, and causal graphical models, with emphasis on exact and parameterized algorithms, counting and sampling, and structural graph methods.

Professional Experience

- Apr 2025–Present** **Institute Postdoctoral Fellow**, Department of Computer Science and Engineering, Indian Institute of Technology Madras, Chennai, India. Joined 9 April 2025. Mentor: Dr. Akanksha Agrawal.
- Aug 2024–Mar 2025** **Senior Project Officer**, Centre for Industrial Consultancy and Sponsored Research, Indian Institute of Technology Madras. Project: *Parameterized Algorithmics–Tradition & Beyond*. Supervisor: Prof. Akanksha Agrawal. Joined 1 August 2024.
- Aug 2014–Jul 2015** **Associate System Engineer**, IBM India Pvt. Ltd., India.

Education

- Aug 2017–Nov 2025** **PhD in Computer Science**, Tata Institute of Fundamental Research, Mumbai, India. Advisor: Prof. Piyush Srivastava. Thesis: *Counting Problems in Graphical Models*. Thesis submitted in January 2025; defended on 31 July 2025; provisional degree certificate issued on 5 August 2025; final degree certificate issued in November 2025.
- Jul 2015–Jun 2017** **M.Tech. in Theoretical Computer Science**, Indian Institute of Technology Guwahati, India. Advisor: Prof. Deepanjan Kesh. Thesis: *Proof of a Non-Trivial Lower Bound in the Bit-Probe Model*.
- Aug 2010–Jul 2014** **B.Tech. in Computer Science and Engineering**, National Institute of Technology Patna, India.

Peer-Reviewed Publications

1. J. Harviainen, P. Parviainen, and **V. S. Sharma**. “Learning Bayesian and Markov Networks with an Unreliable Oracle.” *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*, PMLR 337, 2026.
2. **V. S. Sharma**. “A Fixed-Parameter Tractable Algorithm for Counting Markov Equivalence Classes with the Same Skeleton.” *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024. arXiv:2310.04218.
3. **V. S. Sharma**. “Counting Background Knowledge Consistent Markov Equivalent Directed Acyclic Graphs.” *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023. arXiv:2206.06744.
4. D. Kesh and **V. S. Sharma**. “On the Bitprobe Complexity of Two Probe Adaptive Schemes.” *Discrete Applied Mathematics* 319 (2022), 132–140. Extended abstract in *CALDAM 2019*.

Preprints and Research Notes

- **V. S. Sharma**. “Faster Fixed Parameter Tractable Algorithms for Counting Markov Equivalence Classes with Special Skeletons.” arXiv:2312.17626.
- **V. S. Sharma** and P. Srivastava. “The PSPACE-Hardness of Understanding Neural Circuits.” arXiv:2006.08266.

Selected Research Projects

- Counting and sampling causal structures compatible with observational data and background knowledge, including Markov equivalence classes and causal graphical models with latent variables.
- Parameterized algorithms for graph problems, including minimum consistent subsets, shortest-cycle coverage in planar graphs, biclique contraction, Max-Cut, and structured satisfiability problems.

Conference and Invited Talks

- “Counting Problems in Graphical Models,” invited talk, Modern Trends in Parameterized Algorithms 2025, IIT Madras, December 2025.
- “A Fixed-Parameter Tractable Algorithm for Counting Markov Equivalence Classes with the Same Skeleton,” AAAI Conference on Artificial Intelligence, Vancouver, Canada, February 2024.
- “Counting Background Knowledge Consistent Markov Equivalent Directed Acyclic Graphs,” UAI 2023 (online), August 2023; STCS Annual Talks, TIFR, August 2023; TIFR Open Seminar, June 2023.
- “On the Bitprobe Complexity of Two Probe Adaptive Schemes Storing Two Elements,” CALDAM, IIT Kharagpur, February 2019.

Research Visits and Presentations

- Research visit, University of Bergen, Norway, May–June 2025. Host: Prof. Fedor Fomin.
- “Parameterized Complexity of Minimum Consistent Subset,” poster, IIT Madras Poster Day, February 2026; with Deep Antolia.
- “FPT Algorithms for Shortest Cycle Problems,” poster, IIT Madras Poster Day, February 2026; with Vinod Gupta.

Awards and Research Support

- Microsoft Research Travel Grant, INR 250,000, for presenting at AAAI 2024 in Vancouver, Canada.
- ACM India–IARCS Travel Grant, INR 100,000, for presenting at AAAI 2024 in Vancouver, Canada.

Teaching and Academic Service

- Delivered two lectures in *Topics in Design and Analysis of Algorithms* (CS6100), IIT Madras; instructor: Dr. Akanksha Agrawal.
- Teaching assistant for programming courses during the final three semesters of the M.Tech. programme at IIT Guwahati.
- Reviewer: *Acta Informatica*, AAAI 2026, and NeurIPS 2026.

Student Seminar Talks at TIFR

- Counting Markov equivalence classes with the same skeleton, October 2023.
- Counting Markov equivalence classes with special skeletons, October 2023.
- An introduction to chordal graphs and clique trees, January 2021.
- Integer linear programs and local search for Max-Cut, March 2020.
- Computational complexity of neural network understanding, December 2019.
- Complexity of constant-player potential games, November 2018.

Technical Skills and Languages

Programming: C++, Python. **Languages:** Hindi (native), English (professional proficiency).